Classification of Algebraic Varieties

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July, 2012

Outline of the talk

1. Introduction

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Polynomial equations

Algebraic Geometry is concerned with the study of solutions of polynomial equations say n-equations in m-variables

$$\begin{cases} P_1(x_1,\dots,x_m)=0 \\ \qquad \dots \\ \qquad \dots \\ P_n(x_1,\dots,x_m)=0. \end{cases}$$

1 equation in 1 variable

- For example, 1 equation in 1 variable

$$P(x) = a_d x^d + a_{d-1}x^{d-1} + \ldots + a_1 x + a_0 \qquad a_d \neq 0$$

  (a polynomial of degree d in $\mathbb{C}[x]$).
- If $a_i \in \mathbb{C}$, then by the Fundamental Theorem of Algebra, we may find $c_i \in \mathbb{C}$ such that $P(x)$ factors as

$$P(x) = a_d(x - c_1)(x - c_2) \cdots (x - c_d).$$

- Therefore a polynomial of degree d always has exactly d solutions or roots (when counted with multiplicity).
- If one is interested in solutions that belong to $\mathbb{R}$ (or $\mathbb{Q}$, or $\mathbb{Z}$ etc.), then the problem is much more complicated, but at least we know that there are at most d solutions.

Complex solutions

Throughout this talk I will always look for complex solutions $(z_1, \ldots, z_m) \in \mathbb{C}^m$ to polynomial equations

$$P_1(x_1, \ldots, x_m) = 0 \qquad \ldots \qquad P_n(x_1, \ldots, x_m) = 0$$

where $P_i \in \mathbb{C}[x_1, \ldots, x_m]$.

Projective space

- It is more convenient to work on a compactification $\mathbb{C}^N \subset \mathbb{P}^N_{\mathbb{C}}$ where $\mathbb{P}^N_{\mathbb{C}} = \{\mathbb{C}^{N+1} - (0, \dots, 0)\}/\{\mathbb{C}^*\}$ is projective N-space.
- We have that $\mathbb{P}^0_{\mathbb{C}}$ is a point, $\mathbb{P}^1_{\mathbb{C}} \cong S^2$ is the Riemann sphere and $\mathbb{P}^N_{\mathbb{C}} = \mathbb{C}^N \cup \mathbb{P}^{N-1}_{\mathbb{C}}$.
- We then replace $P(x_1, \dots, x_N)$ by the coresponding homogeneous polynomial in $\mathbb{C}[x_0, \dots, x_N]$. Eg. $x_1^2 + x_2^3 \Rightarrow x_0 x_1^2 + x_2^3$.
- Note that if $P(x_1, \dots, x_N)$ is homogeneous and $\lambda \in \mathbb{C}^*$, then $P(c_1, \dots, c_N) = 0$ iff $P(\lambda c_1, \dots, \lambda c_N) = 0$. Thus the zeroes of $P(x_1, \dots, x_N)$ are a well defined subset of $\mathbb{P}^N_{\mathbb{C}}$.

Bezout's Theorem

Theorem (Bezout's Theorem)

Let $P_1(x_0,\dots,x_n),\dots,P_n(x_0,\dots,x_n)$ be n homogeneous polynomials of degrees $d_1,\dots,d_n$. If the set

$$\bigcap_{i=1}^{n}\{P_i(x_0,\dots,x_n)=0\}\subset\mathbb{P}^n_{\mathbb{C}}$$

is finite, then it consist of exactly $d_1\cdots d_n$ points (counted with multiplicity).

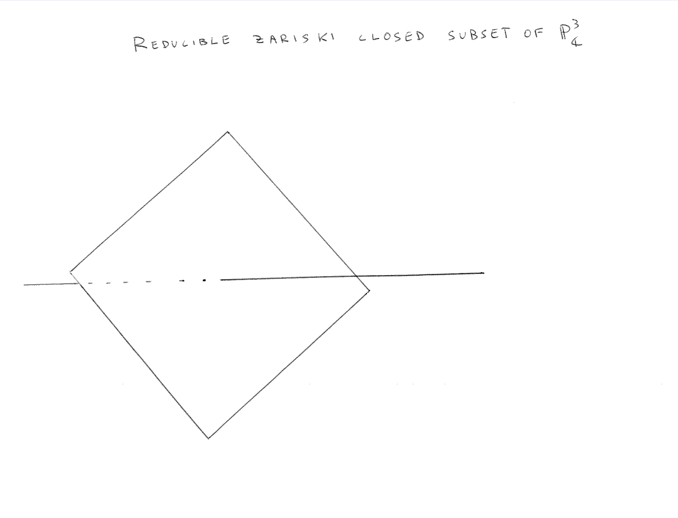
Eg. two distinct lines in $\mathbb{P}^2_{\mathbb{C}}$ always meet in one point.

Outline of the talk

Irreducible subsets

- In general a **Zariski closed subset** of $\mathbb{P}^m_{\mathbb{C}}$ is a set of the form
$$X = \bigcap_{i=1}^{n} \{P_i(x_1, \ldots, x_m) = 0\} \subset \mathbb{P}^m_{\mathbb{C}}.$$
(Where $P_1, \ldots, P_n \in \mathbb{C}[x_0, \ldots, x_m]$ are homogeneous poly's.)
- A Zariski closed subset $X \subset \mathbb{P}^m_{\mathbb{C}}$ is **irreducible** if it can not be written as the union of 2 proper Zariski closed subsets.
- In this case we say that X is a **variety**.
- Any Zariski closed subset is a finite union of irreducible Zariski closed subsets.
- If $P(x, y, z) \in \mathbb{C}[x, y, z]$ is homogeneous of degree $r > 0$, then
$$X = \{P(x, y, z) = 0\} \subset \mathbb{P}^2_{\mathbb{C}}$$
is irreducible if and only if $P(x, y, z)$ does not factor.

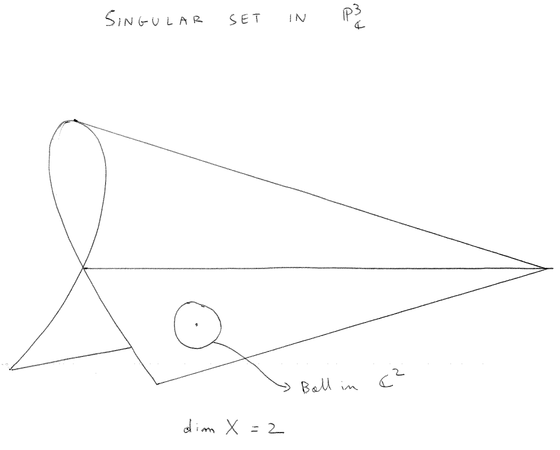
Reducible set



Varieties

- If X is a variety, then there is an integer $d \in \mathbb{N}$ such that for most points $x \in X$, one can find an open neighborhood $x \in U_x \subset X$ where U_x is analytically isomorphic to a ball in $\mathbb{C}^d$.
- We then say that x is a **smooth** point and that X has **dimension** d.
- Most Zariski closed sets are irreducible and smooth.
- By a deep result of Hironaka, there exists a **desingularization** of X, that is a morphism $f : Y \to X$, such that Y is everywhere smooth and f is the identity over any smooth point of X.

Varieties



Outline of the talk

1. Introduction
2. Irreducible subsets
3. Curves
4. Surfaces
5. Higher dimensions

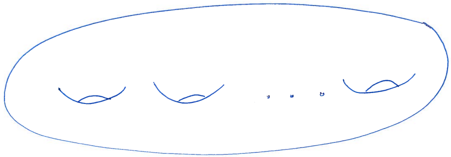
Curves

- A **curve** is a variety of dimension 1.
- All smooth curves are orientable compact manifolds of real dimension 2 and so they are Riemann Surfaces.
- Topologically, a curve X is determined by its genus.
- The simplest curves are **rational curves** i.e. curves of genus 0.
- There is only one such curve $X = \mathbb{P}^1_{\mathbb{C}}$.
- When $g = 1$ we have a 1-parameter family of **elliptic curves** $X_t = \{y^2 - x(x-1)(x-t) = 0\}$.

Curves

$$\mathbb{P}^2_{\mathbb{C}} \supset \{ Y^2 Z^{d-2} = X(X-Z)(X-2Z) \cdots (X-(d-1)Z) \}$$

d even



genus $= \frac{d}{2} - 1$

Curves of general type

- If X is a curve with genus $g \geq 2$ then we say that X is a curve of **general type**.
- These curves share many properties (complicated π_1, negative curvature, finitely many $\mathbb{Q}$-points...).
- There is a $3g-3$-parameter family of curves of general type.
- For any curve X there are (infinitely) many different descriptions

$$\bigcap_{i=1}^{n} \{P_i(x_1, \ldots, x_m) = 0\} \subset \mathbb{P}^m_{\mathbb{C}}.$$

- We would like to find a natural (canonical) description for $X \subset \mathbb{P}^m_{\mathbb{C}}$.

Line bundles

- Since X is compact, the only global holomorphic functions are constant. Therefore, we consider meromorphic functions with prescribed poles or equivalently sections of line bundles.
- If L is a line bundle on X and $s_0, \ldots, s_m$ is a basis of $H^0(X, L)$, then there is a map $\phi_L : X \dashrightarrow \mathbb{P}^m_{\mathbb{C}}$ given by

$$x \to [s_0(x) : \ldots : s_m(x)]$$

(undefined at the common zeroes of the s_i).
- To define an embedding of X in $\mathbb{P}^m_{\mathbb{C}}$, we must find a **very ample** line bundle L (i.e. ϕ_L defines an embedding or equivalently the sections of L distinguish different points and tangent directions).

The canonical line bundle

- There is essentially only one natural choice: the **canonical line bundle** $\omega_X = T_X^\vee$ and its tensor powers $\omega_X^{\otimes n}$.
- The sections of $H^0(X, \omega_X^{\otimes n})$ can locally be written as $f(x)dx^{\otimes n}$.
- We have $\omega_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1}(-2)$ so that $H^0(\omega_{\mathbb{P}^1}^{\otimes n}) = 0$ for $n > 0$.
- When $g(X) = 1$, we have $\omega_X = \mathcal{O}_X$ so that $H^0(\omega_X^{\otimes n}) = \mathbb{C}$ for $n \geq 0$.

Theorem

Let X be a curve of genus $g \geq 2$, then for any $k \geq 3$ the sections of $H^0(X, \omega_X^{\otimes k})$ define an embedding

$$\phi_{\omega_X^{\otimes k}} : X \hookrightarrow \mathbb{P}_{\mathbb{C}}^{(2k-1)(g-1)-1}.$$

The canonical line bundle II

- Setting $k = 3$ and projecting down we have $X \hookrightarrow \mathbb{P}^3_{\mathbb{C}}$ of degree $3(2g-2)$.
- Thus X is defined by at most $3(2g-2)$ equations of degree at most $3(2g-2)$.
- Hence these curves depend on finitely many parameters (in fact on only $3g-3$ parameters).

The canonical ring

- There is a more abstract/natural point of view:
- Define the **canonical ring**

$$R(\omega_X) = \bigoplus_{k \geq 0} H^0(X, \omega_X^{\otimes k}).$$

- If $g = 0$, $R(\omega_X) \cong \mathbb{C}$ and if $g = 1$, $R(\omega_X) \cong \mathbb{C}[t]$.
- If $g \geq 2$, then $\omega_X^{\otimes 3} = \mathcal{O}_{\mathbb{P}^N}(1)|_X$. It follows that $R(X)$ is finitely generated $\mathbb{C}$-algebra which captures the geometry of X. In fact $X = \mathrm{Proj} R(\omega_X)$.
- In coordinates this may be described as follows. If $r_0, \cdots, r_m$ generate $R(\omega_X)$ and $K = \ker(\mathbb{C}[x_0, \ldots, x_m] \to R(\omega_X))$, then $X \subset \mathbb{P}^m_{\mathbb{C}}$ is defined by the equations in K.

Outline of the talk

The canonical ring (in dimension $d \geq 2$)

- Let X be a smooth variety of dimension d, $\omega_X = \wedge^d T_X^\vee$. The sections $s \in H^0(X, \omega_X^{\otimes k})$ may be locally written as $f(x_1, \ldots, x_d)(dx_1 \wedge \ldots \wedge dx_d)^{\otimes k}$.
- The **Canonical Ring** of X is given by

$$R(\omega_X) = \bigoplus_{k \geq 0} H^0(X, \omega_X^{\otimes k}).$$

- The **Kodaira dimension** of X is $\kappa(X) = \mathrm{tr.deg.}_{\mathbb{C}} R(\omega_X) - 1$ (= rate of growth of $\dim H^0(X, \omega_X^{\otimes k})$).
- We have $\kappa(X) = \max_{k \geq 0}\{\dim \phi_{\omega_X^{\otimes k}}(X)\}$ so that $\kappa(X) \in \{-1, 0, 1, \cdots, d\}$. (For $d = 1$ we have $g = 0, 1, \geq 2$, iff $\kappa(X) = -1, 0, 1$.)

Birational geometry of surfaces

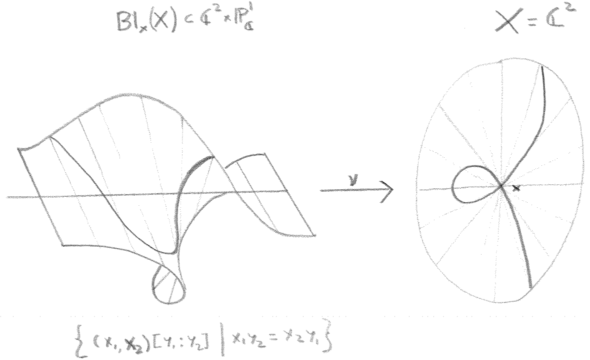
- The classification of (smooth) surfaces ($\dim(X) = 2$) was already understood by the Italian school of Algebraic Geometry around the beginning of the $20^{\rm th}$-century.
- The first problem in the classification of surfaces is that given any point on a surface $x \in X$, one can produce a map

$$\nu : \tilde{X} = \mathrm{Bl}_x(X) \to X$$

known as the **blow up** of X at x.
- The morphism ν may be viewed as a surgery that replaces the point x by a rational curve $E \cong \mathbb{P}^1 = \mathbb{P}(T_xX)$.
- The exceptional curve E is called a -1 **curve** since

$$c_1(\omega_{\tilde{X}}) \cdot E = \deg(\omega_{\tilde{X}}|_E) = -1.$$

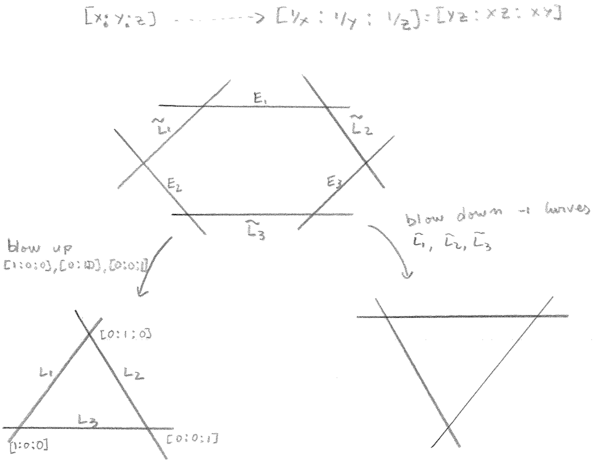
Blow up


$Bl_x(X) \subset \mathbb{C}^2 \times \mathbb{P}^1_{\mathbb{C}}$
$X = \mathbb{C}^2$
ν
$\{(x_1, x_2)[y_1 : y_2] \mid x_1 y_2 = x_2 y_1\}$

Birational geometry of surfaces II

- We say that two surfaces are **birational** if they have isomorphic open subsets (Zariski topology).
- This is a new phenomenon: two curves are birational if and only if they are isomorphic.
- It turns out that two surfaces X and X' are birational if and only if there is a sequence of blow ups $X_n \to X_{n-1} \to \ldots X_1 \to X$ and $X'_m \to X'_{m-1} \to \ldots X'_1 \to X'$ such that $X_n = X'_m$.
- It is then natural to study surfaces up to birational equivalence.

Birational map



Birational geometry of surfaces III

- By a Theorem of Castelnuovo, one may reverse the blowing up procedure i.e. given any surface X and any -1 curve $E \subset X$, there exists a morphism $\nu : X \to X_1$ such that $X = \mathrm{Bl}_x(X_1)$ for some point $x \in X_1$.
- Note that $0 < \rho(X_1) = \rho(X) - 1$ and so this procedure can be repeated at most finitely many times.
- A **minimal surface** is a surface that contains no -1 curves.
- By Castelnuovo's Theorem, given any surface X there exists a birational morphism to a minimal surface $X \to X_{\min}$ (given by finitely many blow downs).
- Note $R(\omega_X) \cong R(\omega_{X_{\min}})$.

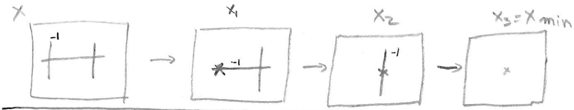
Birational geometry of surfaces IV

- If $\kappa(X) = -1$, then the minimal model $X \to X_{\min}$ is not unique (but it is well understood how two different minimal models are related).
- If $\kappa(X) = -1$, then X is covered by rational curves (i.e. by $\mathbb{P}^1$'s). The sections of $H^0(\omega_X^{\otimes m})$ must vanish along these curves for $m > 0$. Hence $R(\omega_X) \cong \mathbb{C}$.
- If $\kappa(X) \geq 0$, then the minimal model $X \to X_{\min}$ is unique and $\omega_{X_{\min}}$ is **nef**. This means that for any curve $C \subset X_{\min}$ we have
$$c_1(\omega_{X_{\min}}) \cdot C = \deg(\omega_{X_{\min}}|_C) \geq 0.$$
- In fact one can show that $\omega_{X_{\min}}$ is **semiample** (i.e. there is an integer $m > 0$ such that the sections of $H^0(\omega_{X_{\min}}^{\otimes m})$ have no common zeroes).

Birational geometry of surfaces V

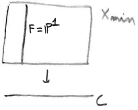
- Therefore, $\phi_{\omega_{X_{\min}}^{\otimes m}} : X_{\min} \to \mathbb{P}^N$ is a morphism, and $\omega_{X_{\min}}^{\otimes m} \cong \phi^* \mathcal{O}_{\mathbb{P}^N}(1)$.
- It then follows that $R(\omega_X)$ is finitely generated.
- If $\kappa(X) = 2$, then $X_{\min} \to X_{\mathrm{can}} = \mathrm{Proj} R(\omega_X)$ is birational (obtained by contracting -2 curves, i.e. rational curves E such that $E^2 = -2$ and $\omega_{X_{\min}} \cdot E = 0$).
- Bombieri shows that $\phi_{\omega_{X_{\min}}^{\otimes 5}}$ is birational and so $X_{\min}$ is birational to a subvariety of $\mathbb{P}^3$ of degree $\leq 25 c_1(\omega_{X_{\min}})^2$.
- Conclusion: If $\kappa(X) = 2$, then we know that if $c_1(\omega_{X_{\min}})^2 \leq M$ then there are only finitely many families of such surfaces (modulo birational isomorphism).

Review



$\kappa(X)<0$

eg $\mathbb{P}^1\times C$

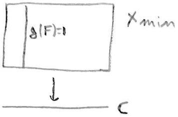


$\kappa(X)=0 \Rightarrow X_{min}$ is Abelian, K3, Enriques or bielliptic surface

eg $E\times E'$

$g(E)=g(E')=1$

$\kappa(X)=1$



eg $F\times C$

$g(F)=1$

$g(C)\geq 2$

$\kappa(X)=2$ eg $C\times C'$ $g(C), g(C')\geq 2$

$$\phi_{\omega_{X_{min}}^{\otimes 5}} : X_{min} \overset{\text{birat}}{\dashrightarrow} \bar{X} \subset \mathbb{P}^3$$

$\uparrow$ deg $\leq c_1(\omega_{X_{min}})^2\cdot 25$

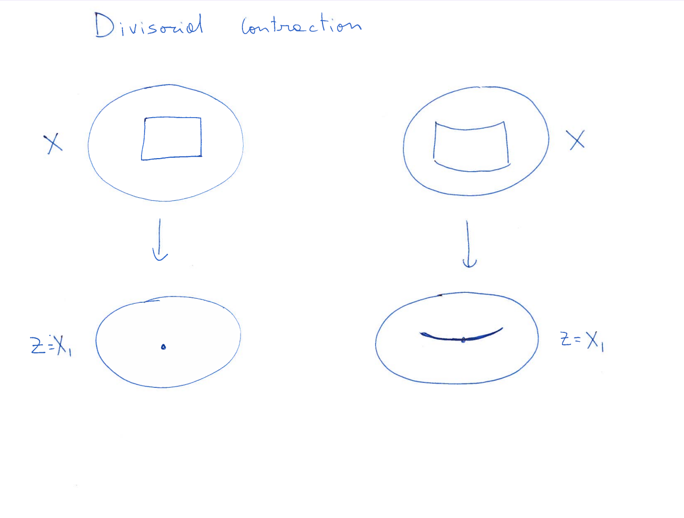
If $c_1(\omega_{X_{min}})^2 \leq M$ then X_{min} belongs to finitely many families

Outline of the talk

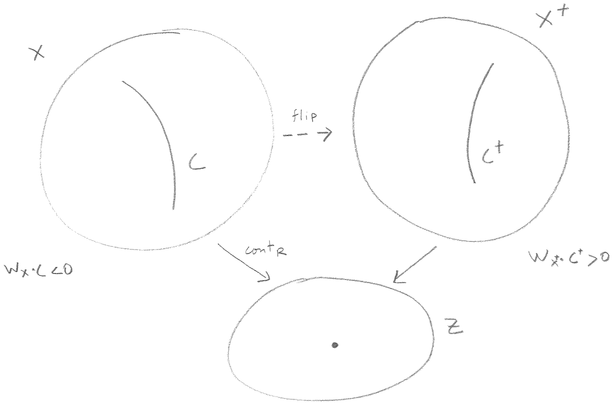
3-folds

- The geometry of complex 3-folds ($d = 3$) was understood in the 1980's by work of S. Mori, Y. Kawamata, J. Kollár, M. Reid, V. Shokurov and others.
- If X is not covered by $\mathbb{P}^1$'s, then there exists a minimal model $X \dashrightarrow X_{\min}$ so that $\omega_{X_{\min}}$ is nef.
- Minimal models can be constructed by finite sequences of well understood birational maps (known as flips and divisorial contractions) $X \dashrightarrow X_1 \dashrightarrow \ldots \dashrightarrow X_k = X_{\min}$.
- Minimal models are not unique (but related by flops).
- $\omega_{X_{\min}}$ is semiample, and so $R(\omega_{X_{\min}})$ is finitely generated.
- If $\kappa(X) = \dim X$, then $X_{\min} \to X_{\mathrm{can}} = \mathrm{Proj} R(\omega_X)$ is birational.
- $X_{\min}$ and X_{can} can be mildly singular.

Divisorial Contraction



Flip



Running the MMP

- The sequence $X \dashrightarrow X_1 \dashrightarrow \ldots \dashrightarrow X_k$ is constructed as follows: If ω_X is nef then X is a minimal model.
- If ω_X is not nef, then (by the Cone Theorem) there is a contraction $\phi : X \to Z$ contracting some curves with $C \cdot c_1(\omega_X) < 0$. (In fact $\rho(X/Z) = 1$.)
- If $\dim Z < \dim X$, then X is covered by $\mathbb{P}^1$s. In this case $\kappa(X) = -1$.
- If $\dim Z = \dim X = \dim \mathrm{Ex}(\phi) + 1$, we say that ϕ is a **divisorial contraction**. In this case Z has mild singularities and we let $X_1 = Z$ and repeat.
- This is the analog of contracting a -1 curve on a surface. In particular $0 < \rho(X_1) = \rho(X) - 1$. Thus we can repeat this operation at most $\rho(X) - 1$ times.

Running the MMP II

- If $\dim Z = \dim X > \dim \mathrm{Ex}(\phi) + 1$, we say that ϕ is a **flipping contraction**. In this case Z is too singular and we must flip ϕ.
- The flip $\phi^+ : X^+ \to Z$ (if it exists) is a map such that $\dim Z = \dim X^+ > \dim \mathrm{Ex}(\phi^+) - 1$ and ϕ^+ contracts curves with $C \cdot \omega_{X^+} > 0$.
- The flip $\phi^+ : X^+ \to Z$ is given by $\mathrm{Proj}_Z R(\omega_X)$.
- So to prove their existence one must show that $R(\omega_X)$ is finitely generated (over Z).
- We can then let $X_1 = X^+$ and repeat the procedure.
- Remark: Note that $\rho(X^+) = \rho(X)$. Never-the-less, it is conjectured that there is no infinite sequence of flips.

Higher dimensions

Theorem (Birkar-Cascini-Hacon-M^{c}Kernan)

Let X be a variety of general type (i.e. $\kappa(X) = \dim(X)$), then there exists a finite sequence of flips and divisorial contractions

$$X \dashrightarrow X_1 \dashrightarrow \dots \dashrightarrow X_k = X_{\min}$$

such that $\omega_{X_{\min}}$ is nef and semiample.

- So we have $\phi^{\otimes m}_{\omega_{X_{\min}}} : X_{\min} \to X_{\rm can} \subset \mathbb{P}^N$ some $m \in \mathbb{N}$.
- The minimal model $X_{\min} = X_k$ has mild singularities but it is not uniquely determined.
- The **canonical model** $X_{\rm can} = \mathrm{Proj}(R(\omega_X))$ is unique, $X_{\rm can}$ has mild singularities and $\omega^{\otimes m}_{X_{\rm can}}$ is very ample (some $m > 0$).

Finite generation

Theorem (Birkar-Cascini-Hacon-M^{c}Kernan, Siu)

The canonical ring

$$R(\omega_X) = \bigoplus_{k \geq 0} H^0(X, \omega_X^{\otimes k})$$

is finitely generated.

It follows that flips exist (but the termination of flips conjecture is open).

Higher dimensional boundedness

One can also generalize some of the features of the classification of surfaces. For example:

Theorem (Tsuji, Hacon-M^{c}Kernan, Takayama)

For any integers $d, M > 0$, there are only finitely many families of varieties of general type, dimension d and volume $\mathrm{Vol}(\omega_X) = c_1(\omega_{X_{\min}})^d \leq M$, modulo birational isomorphisms.

In particular the set $V_d = \{\mathrm{Vol}(\omega_X) \mid \dim X = d\}$ is discrete.

Moduli spaces of canonical polarized varieties

Using techniques of Viehweg, Kollár, Shephered-Barron, Alexeev and others, one can refine the above results to show:

Theorem

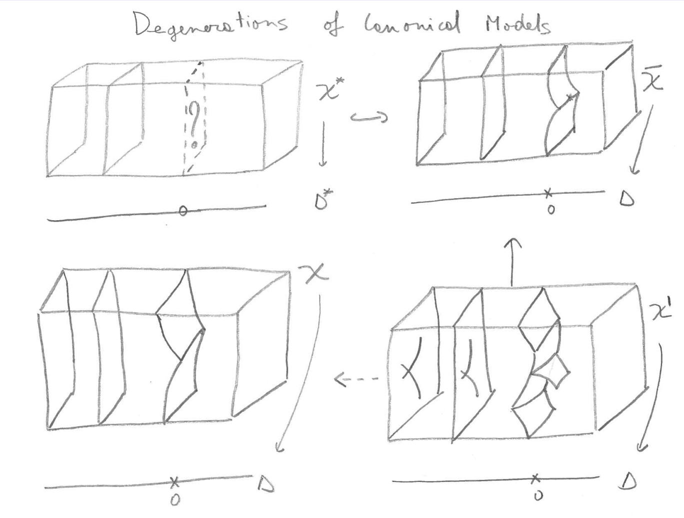
Fix d and M. Then the set of canonical models X_{can} of varieties with $\dim X = \kappa(X) = d$ and $\mathrm{Vol}(\omega_X) \leq M$ admits a coarse moduli space $\mathcal{M}_{d,M}$. In particular, the canonical models X_{can} as above are in 1-1 correspondence with the points of $\mathcal{M}_{d,M}$.

Naturally, one would like to compactify $\mathcal{M}_{d,M}$ in a geometrically meaningful way. In particular one should understand the natural degenerations of canonical models.

Degenerations of canonical models

- Given a family $\mathcal{X}^* \to \Delta^* = \Delta - \{0\}$ of canonical models, consider a compactification $\overline{\mathcal{X}} \to \Delta$.
- $\overline{\mathcal{X}}_0$ may be very singular, so let $\mathcal{X}' \to \overline{\mathcal{X}}$ be a resolution of singularities. Assume that $\mathcal{X}'_0 = Y'_1 \cup \ldots \cup Y'_n$ is a union of smooth codimension 1 subvarieties meeting transversely.
- Let $\mathcal{X} = \mathrm{Proj}_\Delta R(\omega_{\mathcal{X}'})$. Then $\mathcal{X}$ agrees with $\mathcal{X}^*$ over Δ^* and $\mathcal{X}_0 = Y_1 \cup \ldots \cup Y_n$ is SLC (i.e. has mild singularities, e.g. a union of smooth codimension 1 subvarieties meeting transversely).
- This procedure gives us a geometrically meaningful compactification $\mathcal{M}_{d,M} \subset \overline{\mathcal{M}_{d,M}}$.

Degenerations



Moduli spaces of canonical models

- Thus the correct approach is to consider moduli spaces of reducible canonical models $X_{\mathrm{can}} = Y_1 \cup \ldots \cup Y_n$.
- This can be studied by considering irreducible components $Y = Y_i$
$$\omega_{\mathcal{X}}|_Y = \omega_Y(B)$$
where we now allow poles along $B = \sum_{j \neq i}(1 - \frac{1}{n_j})B_j$.
- In particular we allow poles along $Y \cap Y_j \subset Y$ (and other sings).
- New technical issues and difficulties arise when studying the geometry of these pairs (Y, B). However there have been sveral recent breakthroughs. For example we have the following:

Moduli spaces of canonical models

Theorem (Hacon-M^c^Kernan-Xu)

For any integer $d > 0$ the **set of volumes** $\mathrm{Vol}(\omega_X(\sum(1-\frac{1}{n_i}B_i)))$ *where X is smooth,* $\dim X = d$*, $\sum B_i$ is a union of codimension 1 subvarieties, and $n_i \in \mathbb{N}$* **has no accumulation points from above***. In particular there exists $\epsilon > 0$ such that if* $\mathrm{Vol}(\omega_X(\sum(1-\frac{1}{n_i}B_i))) > 0$*, then* $\mathrm{Vol}(\omega_X(\sum(1-\frac{1}{n_i}B_i))) > \epsilon$*. Eg. if* $\dim X = 1$ *then* $\epsilon = 1/42$*.*

It appears that all the pieces are now in place to construct the moduli space of canonical models of SLC pairs (X, B) (thanks to work of Kollár, Abramovich, Alexeev, Hacking, Hacon, Hassett, Karu, Kovács, M^{c}Kernan, Viehweg, Xu and others).